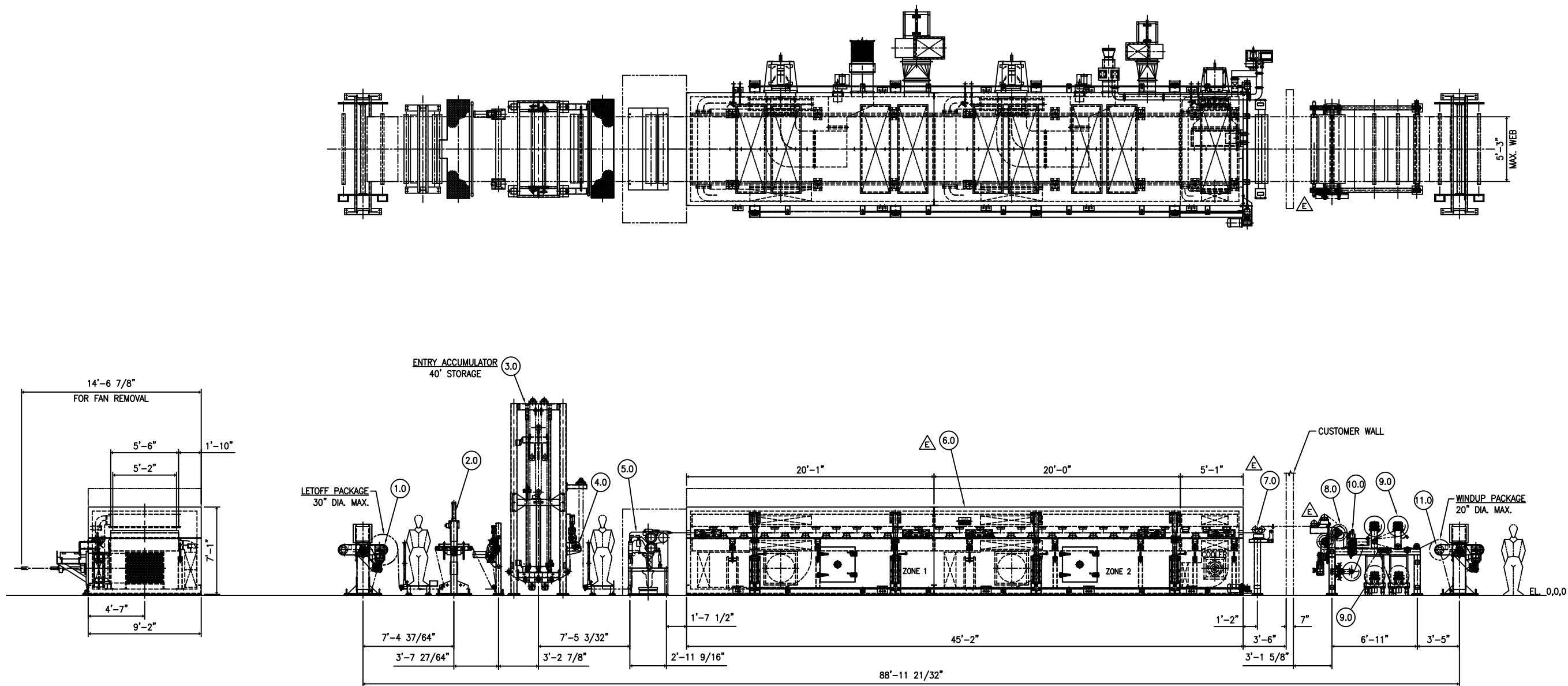


△ TECHNICAL SPECIFICATIONS

PRODUCT: WOVEN, CARBON FIBER, KEVLAR, CARBONIZED RAYON  
WEB WIDTH: 63" MAX.  
FABRIC WEIGHT: 100-600 g/M<sup>2</sup>  
DIP SOLVENT: ACETONE, DMF, MEK, WATER, PM. ALCOHOL  
RESIN: EPOXY, PHENOLIC, POLYESTER, CYANATE ESTER, POLYIMIDE  
LINE SPEED: 5-30 FPM, GEAR-IN FOR 50 FPM ATE REDUCED TENSION  
PROCESS: UNWIND - DIP - DRY - WINDUP  
TENSION: 0.5-2.5 PLI  
OVEN TEMPERATURE: 450°F MAX.  
NOZZLE VELOCITY: 6000 FPM (ADJUSTABLE)  
ELECTRICAL: 460V/3PH/60HZ  
CONTROL: 24V DC  
COMPRESSED AIR: 5.5 BAR  
CHILLED WATER: 50°F @ 2 BAR  
FUEL: NATURAL GAS, 5 PSI @ CONNECTION TO GAS TRAIN

- 1.0 - TURRET LETOFF
- 2.0 - SPLICE PRESS
- 3.0 - ENTRY ACCUMULATOR
- 4.0 - #1 TENSION STAND
- 5.0 - IMPREGNATION STATION
- 6.0 - FLOTATION DRYER
- 7.0 - EDGE CONTROL
- 8.0 - #2 TENSION STAND
- 9.0 - POLY LETOFFS
- 10.0 - SLITTER
- 11.0 - TURRET WINDER



REF. DWG. DS388201G

REVISIONS

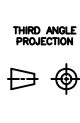
- △ 4/1/08 JAF GENERAL REVISIONS
- △ MRK 6/2/08 ADDED TECHNICAL SPECIFICATIONS INFORMATION AND UPDATED VIEWS OF THE EQUIPMENT

- △ PRS 8/12/08 GENERAL REVISION
- △ UPDATE VIEWS OF ITEMS 6.0, 7.0 AND 8.0 WS 18MAY09



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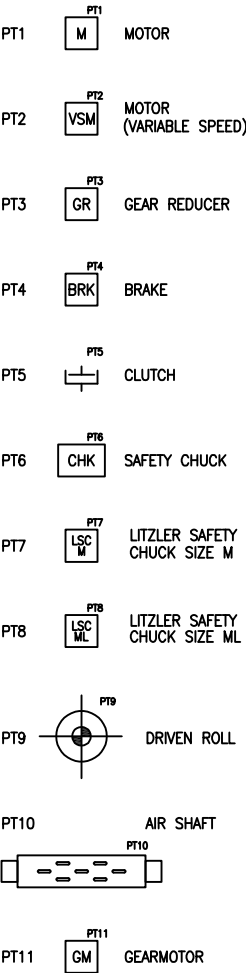
UNLESS OTHERWISE SPECIFIED:  
ALL DIMENSIONS ARE IN INCHES  
ALL DIMENSIONS ARE EXPRESSED IN DECIMALS  
BROKEN ALL SHARP EDGES AND CORNERS  
REMOVE ALL MACHINING BURRS  
ALLOWABLE VARIATIONS:  
FRACTIONS: 0.000 INCHES (0.00 MM) AND DECIMALS: .001 INCHES (0.025 MM)  
ANGLES: 30'-30'

DRAWING APPROVAL  
☐ APPROVED AS DRAWN ☐ APPROVED AS NOTED  
NAME DATE

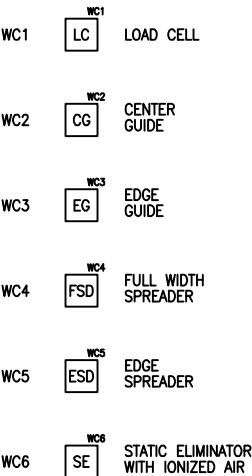
C. A. LITZLER CO., INC.  
CLEVELAND, OHIO 44135-2689, U.S.A. 216-267-8020 (FAX) 216-267-9856  
TITLE DESIGN INFORMATION DOCUMENT  
FOR  
DRAWN BY MRK 2/28/08  
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JOB NUMBER: 1615 SECTION: 98.24  
SCALE: N.T.S.  
DWG. NO. DA1253101E  
SHEET 5  
TOTAL DRAWING SHEETS 5

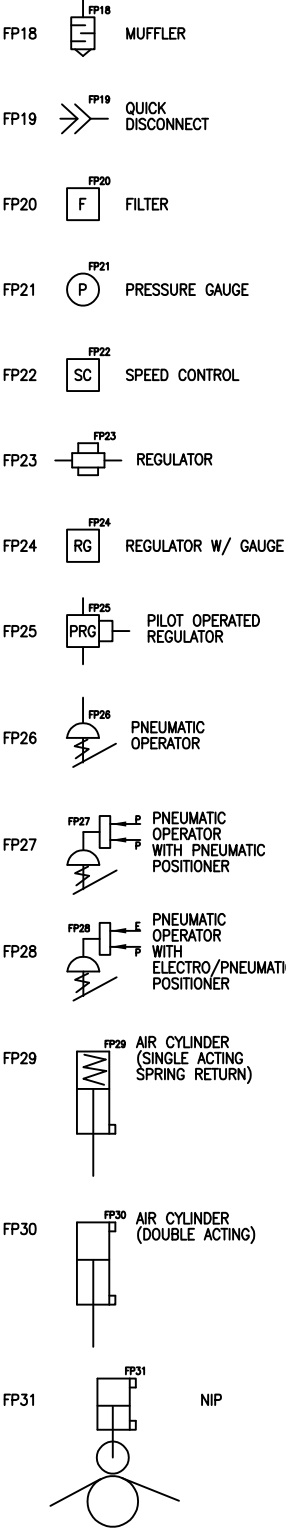
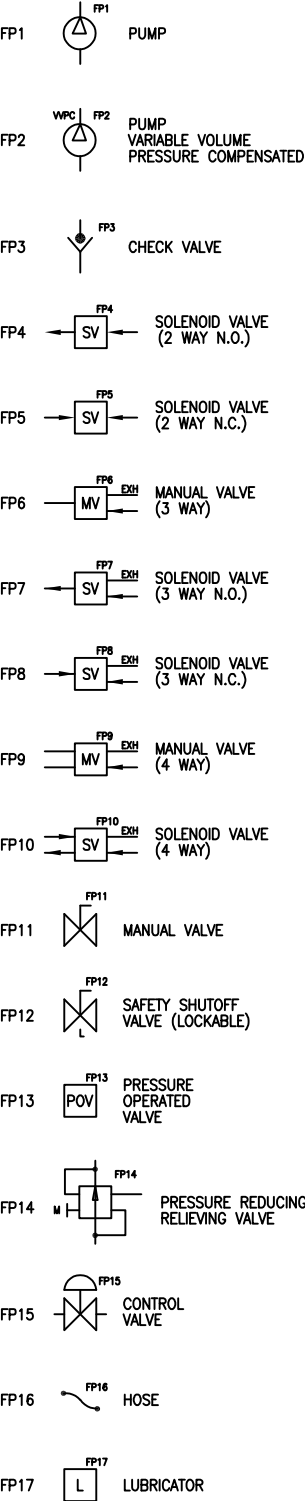
POWER & TRANSMISSION



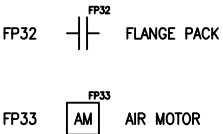
WEB CONTROL



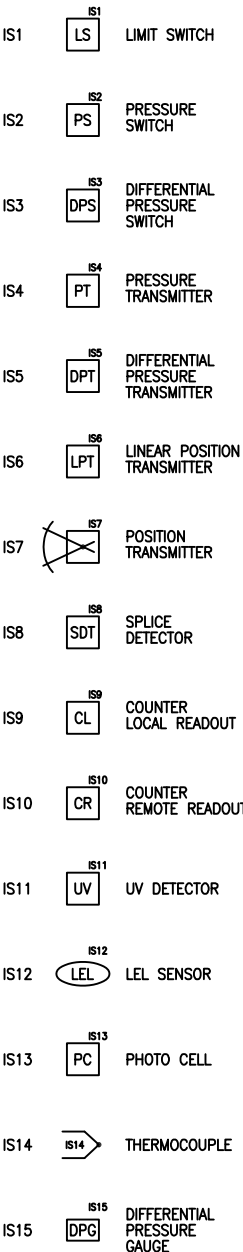
FLUID POWER



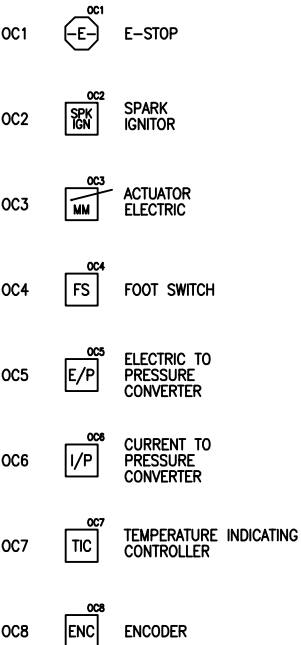
OUTPUT CONTROL



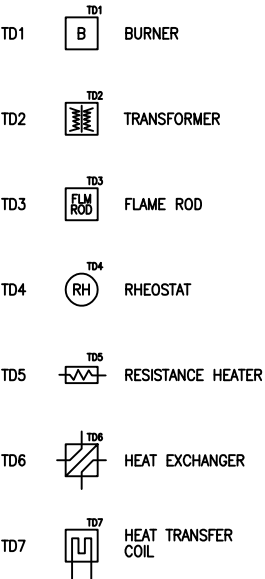
INPUT SENSOR



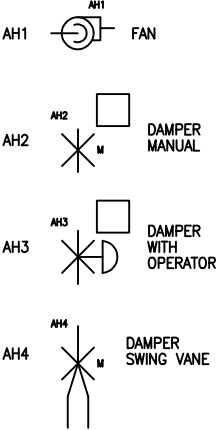
OUTPUT CONTROL



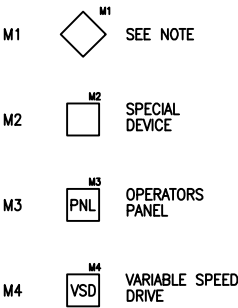
THERMAL DEVICES



AIR HANDLING



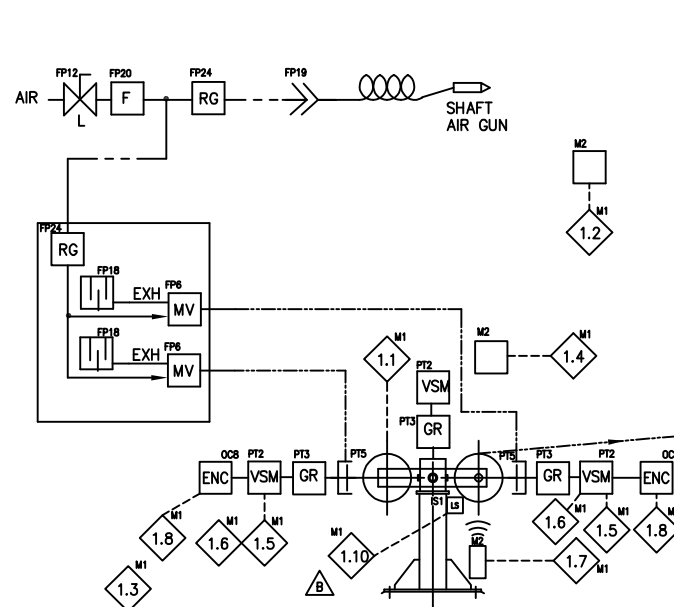
MISCELLANEOUS



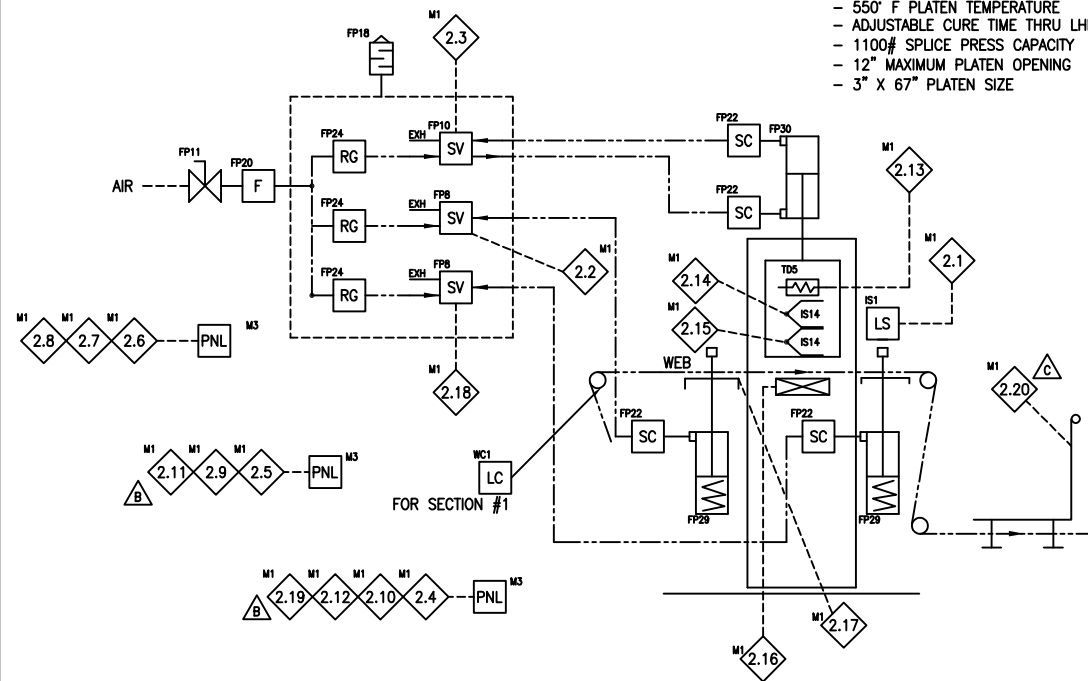
| LINE LEGEND |                      |
|-------------|----------------------|
| .....       | ELECTRICAL (CONTROL) |
| -----       | ELECTRICAL (POWER)   |
| -----       | FLUID (HYD./PNEU.)   |
| -----       | WATER                |
| -----       | PRODUCT PASS LINE    |
| -----       | AIR                  |

## 1.0 DUAL LETOFF

- 1.1 LETOFF PACKAGE:
  - ID. 3" WITH 6" ADAPTERS
  - DIAMETER: 30"
  - WEIGHT: 1500LBS
  - MAX WIDTH: 63"
-  1.2 LETOFF CONTROL PANEL
  - WITH: LETOFF A/B SELECTOR SWITCH
  - LETOFF A & B START-STOP-JOG PUSHBUTTONS
  - E-STOP PUSHBUTTONS
  - YARDAGE COUNTERS
  - TURRET FOR/REV SELECTOR SWITCH
- 1.3 SPINDLE DRIVE MOTOR SPEED SIGNAL TO BE USED WITH MASTER LINE SPEED SIGNAL TO ALARM FOR ACCUMULATOR FILL
  - ALARM W/MANUAL FILL
  - ALARM W/AUTO FILL
  - LHMI ALARM
- 1.4 AIR SHAFT LUG TYPE
- 1.5 LETOFF SPEED CONTROLLED BY LOAD CELL
- 1.6 LETOFF MOTORS, 1000:1 SPEED RANGE
- 1.7 ULTRASONIC SENSOR FOR TENSION CONTROL
- 1.8 ENCODER FEEDBACK TO VFD
-  1.9 NOT USED
-  1.10 LIMIT SWITCH TO INDICATE THE TURRET ARM POSITION FOR TENSION CONTROL



## 1.0 TURRET LETOFF

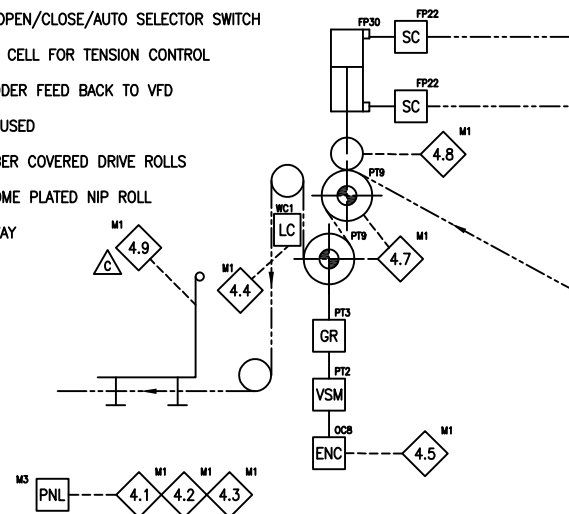


## △ 2.0 SPLICE PRESS

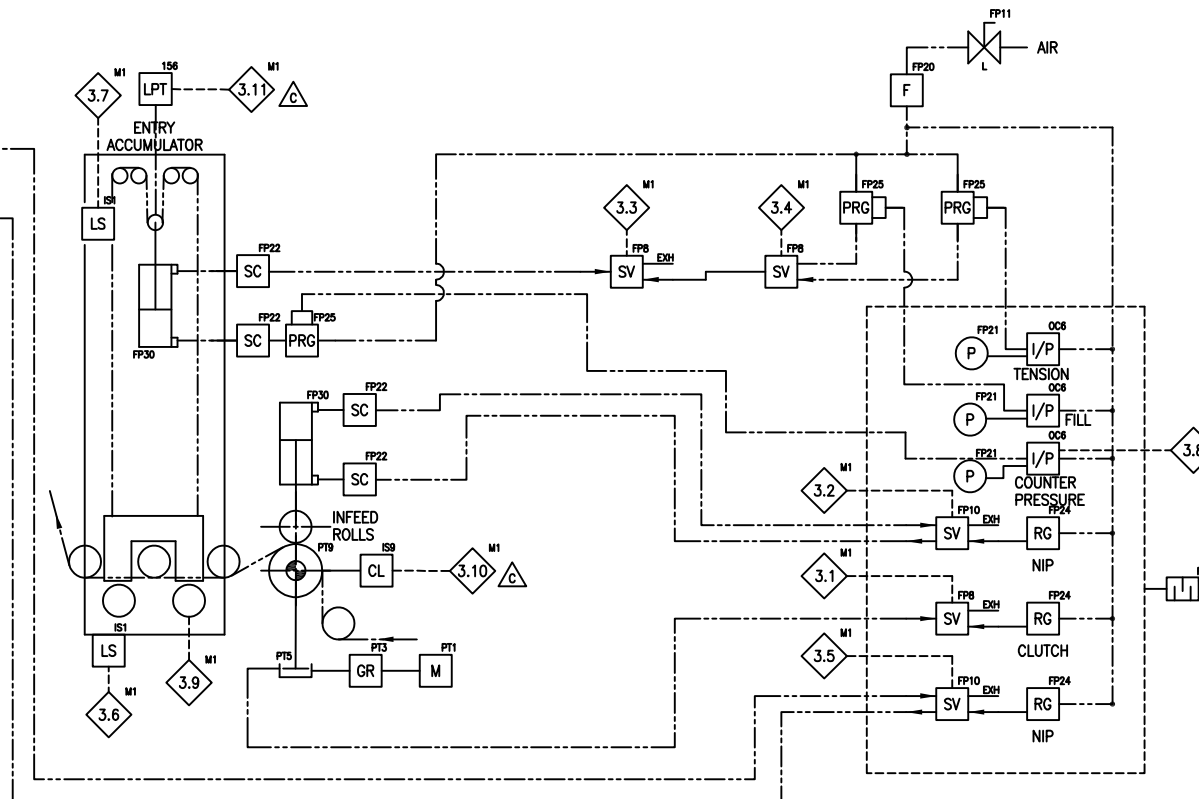
- ## 2.0 SPLICE PRESS
- 2.1 LIMIT SWITCH IS ACTUATED ANY TIME MOVABLE PLATEN IS CLOSED
  - 2.2 SOLENOID VALVE IS ENERGIZED TO CLOSE INLET FABRIC CLAMP
  - 2.3 SOLENOID VALVE IS ENERGIZED TO CLOSE SPLICE PRESS PLATEN
  - 2.4 PUSHBUTTON FOR INLET FABRIC CLAMP
  - 2.5 PUSHBUTTON FOR FEED ROLL FORWARD JOG (REF. SECTION 3.0)
  - 2.6 PUSHBUTTON FOR PLATEN CLOSE (LEFT HAND)
  - 2.7 PUSHBUTTON FOR PLATEN OPEN
  - 2.8 PUSHBUTTON FOR PLATEN CLOSE (RIGHT HAND)
  - 2.9 PUSHBUTTON FOR FEED ROLL REVERSE JOG
  - 2.10 E-STOP PUSHBUTTON
  - 2.11 FEED ROLL NIP PUSHBUTTON (REF. SEC. 3.0)
  - 2.12 FILL ACCUMULATOR PUSHBUTTON
  - 2.13 HEATED UPPER PLATEN WITH PLC CONTROL WITH UPSCALE T/C BREAK PROTECTION. HEATING TIME TO BE PLC CONTROLLED. 
  - 2.14 CONTROL T/C HAS POSITIVE CONTACT WITH PLATEN
  - 2.15 HIGH LIMIT T/C HAS POSITIVE CONTACT WITH PLATEN
  - 2.16 SILICONE RUBBER LOWER PLATEN
  - 2.17 RAZOR GROOVE FOR WEB CROSS CUTTING
  - 2.18 SOLENOID VALVE IS ENERGIZED TO CLOSE EXIT FABRIC CLAMP
  - 2.19 PUSHBUTTON FOR EXIT FABRIC CLAMP
  -  2.20 WALKWAY

#### 4.0 #1 TENSION STAND

- 4.1 DRIVE RUN/JOG FORWARD/JOG REVERSE SELECTOR SWITCH
- 4.2 E-STOP PUSHBUTTON
- 4.3 NIP OPEN/CLOSE/AUTO SELECTOR SWITCH
- 4.4 LOAD CELL FOR TENSION CONTROL
- 4.5 ENCODER FEED BACK TO VFD
- 4.6 NOT USED
- 4.7 RUBBER COVERED DRIVE ROLLS
- 4.8 CHROME PLATED NIP ROLL
- 4.9 WALKWAY
- 



#### 4.0 No. 1 TENSION STAND

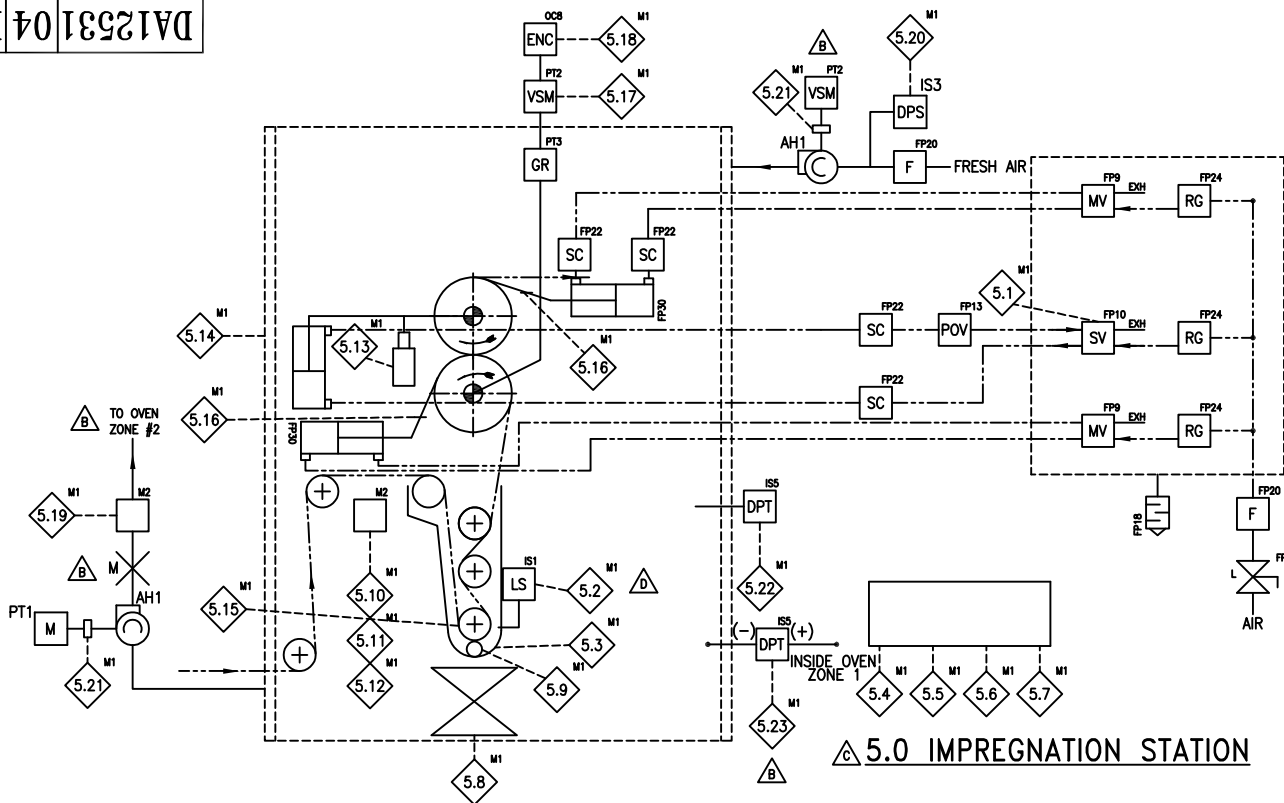


### 3.0 ENTRY ACCUMULATOR AND FEED ROLL

- ### 3.0 ENTRY ACCUMULATOR AND FEED ROLL

- 3.1 SOLENOID VALVE IS ENERGIZED TO ENGAGE CLUTCH
- 3.2 SOLENOID VALVE IS ENERGIZED TO CLOSE FEED ROLL NIP
- 3.3 SOLENOID VALVE IS ENERGIZED TO PRESSURIZE CARRIAGE LIFT CYLINDERS
- 3.4 SOLENOID VALVE IS ENERGIZED TO SELECT HIGH PRESSURE LIFT AIR TO FILL ACCUMULATOR, LOWER PRESSURE AIR IS FOR TENSION CONTROL AS ACCUMULATOR EMPTIES
- 3.5 SOLENOID VALVE IS ENERGIZED TO CLOSE #1 TENSION STAND NIP (REF. SECTION 4.0)
- 3.6 LOWER LIMIT SWITCH INDICATES ACCUMULATOR EMPTY AND REMOVES CARRIAGE LIFT PRESSURE
- 3.7 UPPER LIMIT SWITCH INDICATES ACCUMULATOR FULL AND OPENS THE TENSION STAND NIP
- 3.8 COUNTER PRESSURE USED TO SMOOTH START OF CARRIAGE TRAVEL
- 3.9 DEAD SHAFT STEEL ROLLS WITH CHROME FINISH, 4" DIAMETER
-  3.10 YARDAGE COUNTER WITH LOCAL READOUT
-  3.11 LINEAR TRANSDUCER TO INDICATE THE CARRIAGE POSITION FILL PRESSURE TO BE REGULATED BY CARRIAGE POSITION

STORAGE - 2 MIN @ 20 FPM  
ROLL DIA - 4"  
FEED ROLL DIA - 6" CHROME PLATED  
NIP ROLL DIA - 6" RUBBER COVERED



5.0 IMPREGNATION STATION

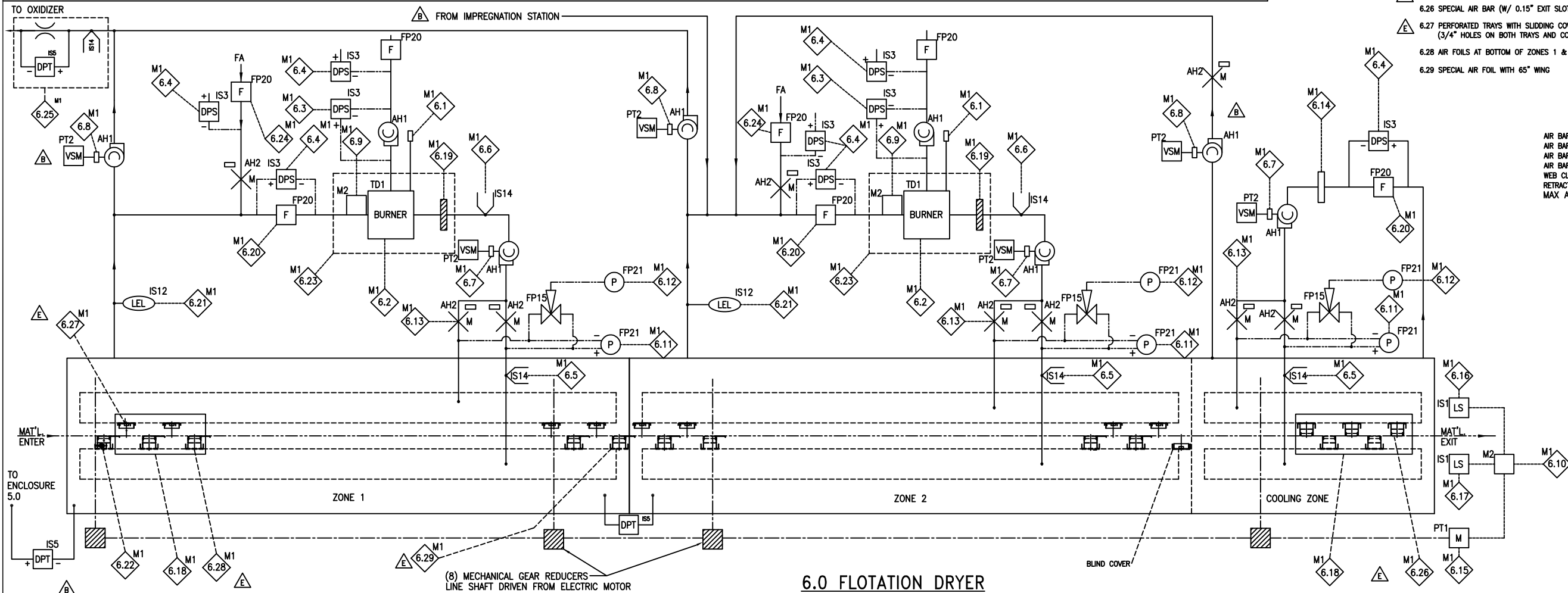
## 5.0 IMPREGNATION STATION

- 5.1 ENERGIZE SOLENOID VALVE TO CLOSE METERING ROLLS
- 5.2 LIMIT SWITCH FOR PAN UP INDICATION
- 5.3 STAINLESS STEEL DIP TANK
- 5.4 SPLICE JUMP PUSHBUTTON
- 5.5 METERING ROLL NIP OPEN/CLOSE SELECTOR SWITCH
- 5.6 E-STOP PUSHBUTTON
- 5.7 METERING ROLL JOG SELECTOR SWITCH
- 5.8 COMMERCIAL LIFT TABLE SYSTEM
- 5.9 2" IPS COUPLING
- 5.10 LOW LEVEL SENSOR IN OVER-FLOW TROUGH
- 5.11 HIGH LEVEL SENSOR
- 5.12 LOW LEVEL SENSOR
- 5.13 SERVO MOTORS FOR GAP ADJUSTMENT
- 5.14 ENCLOSURE
- 5.15 (3) 3" CHROME PLATED DIP ROLLS
- 5.16 DOCTOR BLADE HOLDERS
- 5.17 MOTORS TO BE EXPLOSION PROOF
- 5.18 ENCODER FEED BACK TO VFD
- 5.19 LEL SENSOR FOR AREA MONITORING
- 5.20 FILTER DIFFERENTIAL PRESSURE SWITCH (0.4"-1.6")
- 5.21 FAN ROTATION SWITCH
- 5.22 DIFFERENTIAL PRESSURE TRANSMITTER TO MONITOR ENCLOSURE PRESSURE AND TO CONTROL SUPPLY FAN SPEED
- 5.23 DIFFERENTIAL PRESSURE MONITOR

## 6.0 FLOTATION DRYER

- 6.1 F.M. GAS TRAIN WITH DIRECT SPARK IGNITION AND FLAME ROD
- 6.2 MAXON OVEN PACK BURNER, 1.55 MIL BTU/HR
- 6.3 COMBUSTION BLOWER DIFFERENTIAL SWITCH (0.15"-0.55")
- 6.4 FILTER DIFFERENTIAL PRESSURE SWITCH (0.4"-1.6")
- 6.5 CONTROL THERMOCOUPLE (DUAL TYPE)
- 6.6 HIGH LIMIT THERMOCOUPLE
- 6.7 RECIRCULATING FAN ROTATION SWITCH
- 6.8 EXHAUST FAN ROTATION SWITCH
- 6.9 BURNER OBSERVATION PORT
- 6.10 RETRACTION CONTROL PANEL, NON-Classified LOCATION AT EXIT END OF COOLING ZONE (MOMENTARY OPERATION SELECTOR SWITCH) WITH ELECTRONIC HOLDER FOR RETRACTION PINS
- 6.11 UPPER/LOWER HEADER DIFFERENTIAL PRESSURE GAUGE (-2" TO +2")
- 6.12 TOTAL PLENUM PRESSURE GAUGE (0" TO 10")
- 6.13 PLENUM TOP AND BOTTOR BALANCE DAMPERS (TO ADJUST TOP AIR VELOCITY RELATIVE TO THE BOTTOM VELOCITY) MUST OPERATE FROM OUTSIDE THE OVEN
- 6.14 COOLING COIL
- 6.15 RETRACTION MOTOR
- 6.16 RETRACTION UP LIMIT SWITCH (STOPS RETRACTION MOTOR)
- 6.17 RETRACTION DOWN LIMIT SWITCH (STOPS RETRACTION MOTOR)
- 6.18 VIEWING DOORS
- 6.19 MIXING SCREEN
- 6.20 RECIRCULATION AIR FILTER
- 6.21 LEL SENSORS-SNR674
- 6.22 ENTRY AIR BARS TO BE DAMPERED
- 6.23 304 STAINLESS STEEL INTERNAL HEATER/MIXER BOX
- 6.24 FRESH AIR FILTER BY CUSTOMER
- 6.25 MASS FLOW INDICATOR ON HMI. (SCFM)
- 6.26 SPECIAL AIR BAR (W/ 0.15" EXIT SLOT, 0.75 INNER SLOT)
- 6.27 PERFORATED TRAYS WITH SLIDING COVER ON TOP OF ZONES 1 & 2 (3/4" HOLES ON BOTH TRAYS AND COVERS)
- 6.28 AIR FOILS AT BOTTOM OF ZONES 1 & 2.
- 6.29 SPECIAL AIR FOIL WITH 65" WING

AIR BAR LENGTH: 66"  
AIR BAR PITCH: 20"  
AIR BAR VELOCITY: 6000 FPM (ADJUSTABLE)  
AIR BAR MAT'L: STAINLESS STEEL  
WEB CLEARANCE: 0-0.5"  
RETRACTION: 18"  
MAX AIR TEMP: 450°F



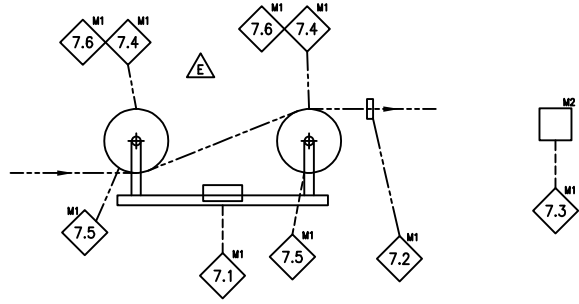
6.0 FLOTATION DRYER

REF. DWG. DA976401

|                                                                                                                   |                                                                             |                                                                                    |                                                 |                                             |                                                                                                                                                                                                        |                               |                                                                                                                                                                                                                                                                                                                |                                                                     |                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>4/1/08 JAF<br/>ADDED CASCADE EXHAUST FOR COOLER TO ZONE #2, IMPREGNATION EXHAUST USED FOR ZONE #2 MAKE UP.</p> | <p>MRK 6/2/08<br/>CHANGED VIEW OF IMPREGNATION STATION AND CHANGED 5.22</p> | <p>ADD NOTES 6.26, 6.27, 6.28, &amp; 6.29 CORRECT VIEWS TO AS BUILT WS 18MAY09</p> | <p>MCT<br/>MOTOR &amp; CONTROL TECHNOLOGIES</p> | <p>PSC<br/>PNEUMATIC SYSTEMS BY LITZLER</p> | <p>THIS DRAWING IS THE PROPERTY OF THE C. A. LITZLER CO., INC. AND IS LOANED SUBJECT TO RETURN UPON DEMAND. BORROWER AGREES THAT SAME WILL NOT BE USED IN ANY WAY DETRIMENTAL TO OWNER'S INTEREST.</p> | <p>THIRD ANGLE PROJECTION</p> | <p>UNLESS OTHERWISE SPECIFIED<br/>ALL DIMENSIONS ARE IN INCHES<br/>ALL DIMENSIONS ARE EXPRESSED IN DECIMALS<br/>BREAK ALL SHARP EDGES AND CORNERS REMOVE ALL MACHINING MARKS<br/>ALLOWABLE VARIATIONS<br/>FRACTIONS: 1/16" INCHES (1.56 MM)<br/>DECIMALS: .01" = 0.02 INCHES (0.51 MM)<br/>ANGLES: 45°-30°</p> | <p>DRAWING APPROVAL<br/>APPROVED AS DRAWN<br/>APPROVED AS NOTED</p> | <p>C. A. LITZLER CO., INC.<br/>CLEVELAND, OHIO 44135-2689, U.S.A. 216-267-8020 (FAX) 216-267-9856<br/>TITLE: DESIGN INFORMATION<br/>FOR: DRAWN BY: MRK 2/28/08<br/>CHECKED BY: APPROVED BY: APPROVED BY:<br/>JOB NUMBER: 1615 SECTION: 98.24<br/>SCALE: FULL SIZE<br/>DWG. NO. DA12531 04 E<br/>TOTAL DRAWING SHEETS 5</p> |
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### 7.0 EDGE CONTROL

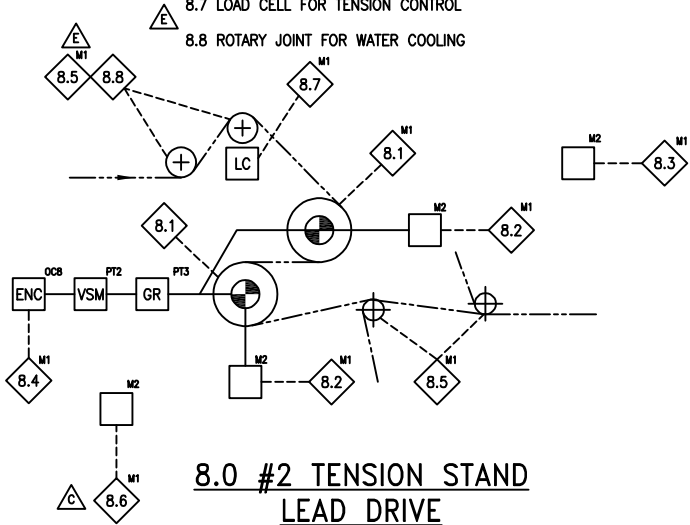
- 7.1 STEERING FRAME WEB GUIDING SYSTEM
- 7.2 EDGE SENSOR
- 7.3 PANEL FOR THE GUIDER
- 7.4 ROLLS TO BE RELEASE COATED
- 7.5 ROTARY JOINT FOR WATER COOLING
- 7.6 ROLLS TO BE 6"



7.0 EDGE CONTROL

### 8.0 #2 TENSION STAND

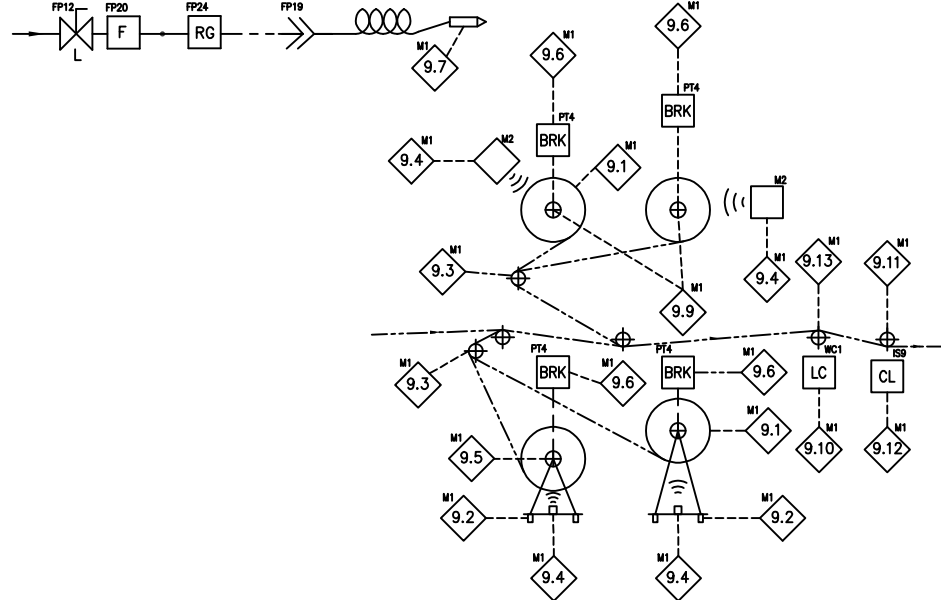
- 8.1 ROLLS ARE PLASMA COATED
- 8.2 ROTARY UNIONS WITH STAINLESS STEEL SYPHON TUBE
- 8.3 ADVANCED COOLING SYSTEM FOR ROLL COOLING
- 8.4 ENCODER FEEDBACK TO VFD
- 8.5 IDLER ROLL TO BE 6" AND RELEASE COATED
- 8.6 TENSION STAND CONTROL PANEL
  - RUN/JOG FORWARD/JOB REVERSE SELECTOR SWITCH
  - E-STOP PUSH BUTTON
- 8.7 LOAD CELL FOR TENSION CONTROL
- 8.8 ROTARY JOINT FOR WATER COOLING



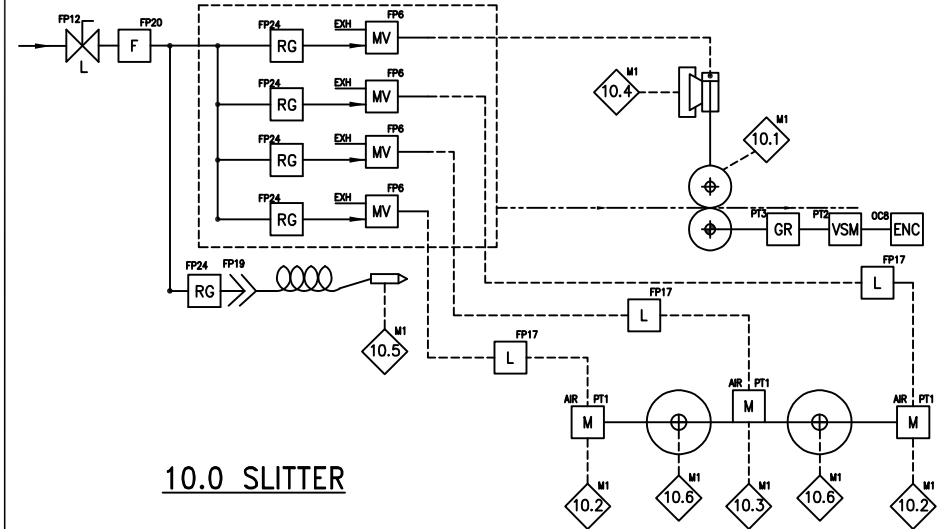
8.0 #2 TENSION STAND LEAD DRIVE

### 9.0 POLY LETOFFS

- 9.1 POLY PACKAGE
  - CORE: 3"
  - MAX DIA: 20"
  - MAX WEIGHT: 200LB.
  - TENSION: 31.5 LB (0.5 PLI) MAX.
- 9.2 LOWER LETOFF ON A CART
- 9.3 EXPANDER ROLL TO SPREAD THE POLY PRIOR TO APPLICATION
- 9.4 ULTRASONIC SENSOR FOR TENSION CONTROL
  - TENSION RANGE: 0.5 PLI.
- 9.5 AIR SHAFT--(4) TOTAL REQUIRED, WITH 1-1/4" SQUARE SHAFT
- 9.6 MAGPOWR MAGNETIC PARTICLE BRAKE
- 9.7 SHAFT INFLATION AIR GUN
- 9.8 NOT USED
- 9.9 UPPER LETOFF ON PLATFORM
- 9.10 LOAD CELLS FOR WINDER TENSION CONTROL
  - CONNECT TO COMMON AMPLIFIER LOCATED NEAR EQUIPMENT
- 9.11 LIVE SHAFT ROLL FOR MOUNTING YARDAGE COUNTER
- 9.12 YARDAGE COUNTER WITH LOCAL READOUT
- 9.13 IDLER ROLLS TO BE 6" WITH RELEASE COATING



9.0 POLY LETOFFS



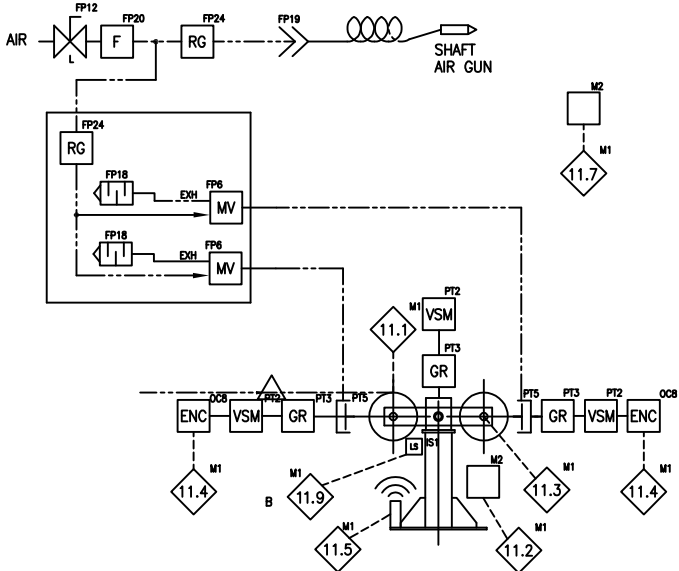
### 10.0 SLITTER

- 10.1 (2) KNIFE OVER ROLL SLITTERS FOR EDGE SLITTING
- 10.2 AIR MOTORS FOR SCRAP WINDERS (2)
- 10.3 AIR MOTOR FOR SCRAP WINDER OSCILATING (1)
- 10.4 SLITTERS ARE MOUNTED ON DOVETAIL BARS TO ALLOW MANUAL POSITIONING
- 10.5 SHAFT INFLATION AIR GUN
- 10.6 AIR SHAFT (LUG TYPE) ON SCRAP WINDERS

10.0 SLITTER

### 11.0 TURRET WINDUP

- 11.1 WINDER PACKAGE:
  - I.D.: 3" WITH 6" ADAPTERS
  - MAX DIA: 20"
  - MAX WEIGHT: 900LB
  - MAX WIDTH: 63"
- 11.2 ROPE SAFETY SWITCH
- 11.3 AIR SHAFT ALUMINUM, LUG TYPE (2)
- 11.4 ENCODER FEEDBACK TO VFD
- 11.5 ULTRASONIC SENSOR FOR TENSION CONTROL
- 11.6 NOT USED
- 11.7 WINDUP CONTROL PANEL
  - TURRET FORWARD/REVERSE SELECTOR SWITCH
  - WINDER A/B SELECTOR SWITCH
  - WINDER RUN/OFF/JOG SELECTOR SWITCH
  - E-STOP PUSHBUTTON
  - YARDAGE COUNTERS
- 11.8 NOT USED
- 11.9 LIMIT SWITCH TO INDICATE THE TURRET ARM POSITION FOR TENSION CONTROL



11.0 TURRET WINDUP

### 13.2 ADVANCED COOLING SYS.

- 13.1 WATER CHILLER BY CUSTOMER.
- 13.2 PRESSURE TRANSDUCER FOR WATER PRESSURE CONTROL.
- 13.3 CHILLED WATER PRESSURE CONTROL.
- 13.4 COOLING COIL CONTROL VALVE.
- 13.5 CONTROL VALVE FOR WATER TEMPERATURE CONTROL IN RECIRCULATION LOOP FOR GUIDER ROLLS AND PULL OUT ROLLS.

